Answer key: Lesson 5, Make-a-Ten Adding

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the ten-frames. Fill in the blanks.

Answer: 1 more; 10 and 3; 9 + 4 = 13

The first frame has 1 empty space, so 9 needs 1 more. Move 1 counter from the 4. The first frame is full (10) and 3 are left. 10 + 3 = 13.

Watch for: Writing 10 and 4: forgetting to take the moved counter away from the 4.; Counting all the counters again and slipping (12 or 14).

S2. Draw 7 dots and 6 dots in the box. Ring 10 dots. Then fill in the blanks.

Answer: 7 + 6 = 10 + 3 = 13 (ten dots ringed, 3 left over)

Ring the 7 dots and 3 of the 6 dots to make ten. The other 3 dots are outside the ring. So 7 + 6 = 10 + 3 = 13.

Watch for: Ringing ten dots but forgetting the dots left outside.; Miscounting when counting every dot one by one.

S3. Fill in the missing numbers to make the next ten.

Answer: 3, 2, 4, 1

(a) 7 and 3 make 10. (b) 18 is 1 ten and 8 ones, and 8 and 2 make 10, so 18 + 2 = 20. (c) 6 and 4 make 10, so 26 + 4 = 30. (d) 9 and 1 make 10, so 39 + 1 = 40.

Watch for: Writing the next ten itself (20 for part b).; Counting from the tens digit instead of the ones digit.

S4. Add 28 + 6. Fill in the blanks.

Answer: 2 more; 6 = 2 and 4; 28 + 6 = 34

28 needs 2 more to make 30. Split 6 into 2 and 4. 28 + 2 = 30, and 30 + 4 = 34.

Watch for: Writing 36: adding all 6 after reaching 30, when 2 of the 6 were already used.; Splitting 6 into 2 and 3 or another wrong pair.

S5. Add 35 + 7. Fill in the blanks.

Answer: 5 more; 35 + 7 = 42

35 needs 5 more to make 40. 7 is 5 and 2. 35 + 5 = 40, and 40 + 2 = 42.

Watch for: Writing 47 (adding 7 to the 4 tens).; Forgetting the 2 left over after the ten.

S6. Add.

Answer: 15, 23, 32, 3 more, 53

(a) 9 + 1 = 10, 10 + 5 = 15. (b) 18 + 2 = 20, 20 + 3 = 23. (c) 27 + 3 = 30, 30 + 2 = 32. (d) 47 needs 3 more to make 50, 6 is 3 and 3, so 47 + 3 = 50 and 50 + 3 = 53.

Watch for: Moving the right amount to make the ten but then adding the whole second number.; Making a ten with the wrong number (for example 9 + 6 = 9 + 4 + 2).

S7. Siti adds 8 + 6. Here is her working: 8 + 2 = 10, then 10 + 6 = 16. Answer (a), (b) and (c).

Answer: Wrong. The right answer is 14. 10 + 6 should be 10 + 4: she moved 2 from the 6, so only 4 are left.

8 + 2 = 10 uses 2 of the 6. Only 4 are left, so the sum is 10 + 4 = 14. Siti added all 6 again.

Watch for: Saying she is right because 10 + 6 looks like a good sum.; Writing 14 without saying what she forgot (a spoken reason is fine).

Think harder (page 3)

T1. Mei adds 4 + 9 + 6. Fill in the blanks.

Answer: 4 and 6 make 10. The total is 19.

4 and 6 make 10. 10 + 9 = 19. Check the long way: 4 + 9 = 13, and 13 + 6 = 19.

Watch for: Adding the first two numbers (4 + 9) and stopping or slipping.; Thinking the pair must sit side by side.

T2. Add 8 + 5 + 2 + 7. Show your working on the lines.

Answer: 22 (8 + 2 = 10, 5 + 7 = 12, 10 + 12 = 22)

8 + 2 = 10. 5 + 7 = 12 (5 + 5 + 2). 10 + 12 = 22.

Watch for: Adding from left to right and losing track.; Using a number twice or leaving one out.

T3. Circle the sum that does not make 30. Then write how you know.

Answer: 19 + 12 (it makes 31). The ones 9 and 2 make 11, not 10.

7 + 3 = 10, 8 + 2 = 10 and 6 + 4 = 10, so those make 30. 9 + 2 = 11, so 19 + 12 is 31.

Watch for: Adding every sum one by one and slipping.; Choosing 18 + 12 because 18 looks big.

T4. Ben has 18 red marbles, 34 blue marbles and 12 green marbles. How many marbles does Ben have altogether? Show your working on the lines.

Answer: 64 (18 + 12 = 30, 30 + 34 = 64)

18 + 12 = 30. 30 + 34 = 64. The order does not change the total.

Watch for: Leaving out one group of marbles.; Adding 18 + 34 first and slipping on the ten.

Go further (page 4)

G1. Add 18 + 7 + 12 + 23 + 3 + 17. Then change one number. Show your working on the lines.

Answer: (a) 80 (18 + 12 = 30, 7 + 3 = 10, 23 + 17 = 40). (b) 13: 80 + 10 = 90, so the 3 changes to 13.

(a) 18 + 12 = 30. 7 + 3 = 10. 23 + 17 = 40. 30 + 10 + 40 = 80. Another way: 7 + 23 = 30, 18 + 12 = 30, 3 + 17 = 20, total 80.

(b) The total must go up from 80 to 90, which is 10 more. The 3 must become 3 + 10 = 13.

Watch for: Adding left to right and losing the running total.; Using a number twice or leaving one out.; Changing the 3 to 10 (the total goes up by 7, not 10).

G2. Look at this sum: 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1. Fill in a and b.

Answer: 45 and 55 (four pairs of 10, and the 5 is left)

1 + 9, 2 + 8, 3 + 7 and 4 + 6 make four tens: 40. The 5 has no partner. 40 + 5 = 45. Adding 10 makes 55.

Watch for: Forgetting the 5 in the middle has no partner.; Making only three pairs.

G3. Write a sum of 6 numbers that makes 100. Many answers are right. Show your working on the lines.

Answer: Any six numbers that total 100. For example 45 + 5 + 20 + 17 + 3 + 10 = 100 (45 + 5 = 50, 17 + 3 = 20, 20 + 10 = 30).

Choose pairs first: 50 + 20 + 30 = 100. Then split each into two numbers, for example 45 + 5, 17 + 3, 20 + 10. Add them to check.

Watch for: Choosing numbers that total 90 or 110: add every number to check.; Using fewer than 6 numbers.

Investigation (page 5): Move the fewest

Fill up the bigger number: it has the fewest empty spaces, so you move the fewest dots. The number moved is 10 minus the bigger number. 9 + 6: move 1, not 4. 8 + 7: move 2, not 3. If the two numbers are the same (7 + 7), either way moves 3.

Look for: A child who tries several pairs, records how many dots moved each way, and notices the bigger number is closer to ten. Credit any clear reason, such as the bigger number has fewer empty spaces.

Exit ticket (page 5)

E1. Add.

Answer: 13

7 + 3 = 10. 6 is 3 and 3, so 10 + 3 = 13.

Watch for: Writing 16 (10 + 6).; Writing 12 or 14 after counting on fingers.

E2. Add.

Answer: 43

38 + 2 = 40. 5 is 2 and 3, so 40 + 3 = 43.

Watch for: Writing 45 (40 + 5).; Writing 33 or 83.

E3. Add 6 + 9 + 4. Fill in the blanks.

Answer: 19. 6 and 4 make 10, then 10 + 9 = 19.

6 + 4 = 10. 10 + 9 = 19.

Watch for: Writing 6 and 9 as the pair that makes 10 (6 + 9 is 15).; Not noticing that 6 and 4 make a ten.

Sneak-peek puzzles (not in the workbook)

1. 12
2. Yes. Nothing was added or taken away. 9 + 5 = 14 and 10 + 4 = 14.